

I B.Tech I Semester Regular & Supplementary Examinations, Dec- 2025 / Jan- 2026

Linear Algebra and Calculus

(Common to EEE, ECE, EBM, IT, CSE(AI), CSE(CS), CSE(IOT), CSBS, AI & AIDS)

(For 2023, 2024 & 2025 Admitted Batches Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) Find the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 2 & 3 & 6 \end{bmatrix}$
- 1.b) State Cauchy-Binet Formula.
- 1.c) Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$
- 1.d) Define Eigen vector.
- 1.e) State Rolle's Theorem.
- 1.f) Is the Rolle's theorem applicable to $f(x) = x^2$ in $[1,2]$
- 1.g) Find the stationary points of $f(x, y) = x^2 + y^2 + 6x + 30$
- 1.h) If $u = 2x^2 + 3y^2$ then find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$
- 1.i) Define double integral.
- 1.j) Evaluate $\int_0^2 \int_0^3 (x + y) dx dy$

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Find the rank of a matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -2 & -3 & 1 & 2 \\ -3 & -4 & 5 & 8 \\ 1 & 3 & 10 & 14 \end{bmatrix}$ using Echelon form.
(OR)
- b) Test for consistency and hence solve the system of equations
 $x + y + z = 6, 2x + 3y - 2z = 2, 5x + y + 2z = 13.$
- 3 a) Find Eigen values and Eigen vectors for $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$
(OR)
- b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & -2 & 2 \\ 1 & -2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$
- 4 a) Verify Rolle's theorem for the function $f(x) = \tan x$ in $[0, \pi]$
(OR)
- b) Verify Cauchy's mean value theorem for $f(x) = e^x, g(x) = e^{-x}$ in $[3, 7]$
- 5 a) Find the extremum of $f(x, y) = x^3 + y^3 - 3axy.$
(OR)
- b) Find the point on the plane $x + 2y + 3z = 4$ that is closest to origin.
- 6 a) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by Changing in to polar variables
(OR)
- b) Change the order of integration $\int_0^1 \int_{x^2}^x xy dx dy$

1
0
0
-4
-10
-19
-57

Code: 23AHS05

R23

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Engineering Physics

(Common to EEE, ECE, ECE(BM), IT, CSE(AI), CSE(CS), CSE(IOT), CSE(BS), AI & AIDS)

(For 2023, 2024 & 2025 Admitted Batches Only)

Max.Marks:70

Time: 3 Hours

PART-A

10 x 2 = 20M

Answer all questions and all questions carry equal marks

- 1.a) Mention the necessary conditions for sustained interference of light.
- 1.b) Mention any two applications of newton's rings.
- 1.c) What is the coordination number of a crystal structure?
- 1.d) State Bragg's law of diffraction.
- 1.e) Brief the properties of "wave function".
- 1.f) Brief "Quantum free electron theory".
- 1.g) Define forbidden energy gap in semiconductors.
- 1.h) List any four properties of superconductors.
- 1.i) Define "Dielectric Constant".
- 1.j) Define "Hysteresis loss".

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Discuss Fraunhofer diffraction due to double slit with a neat diagram.

(OR)

- b) Discuss the dispersive and resolving power of grating in detail.

- 3 a) Define Space Lattice. Distinguish between polycrystalline and amorphous materials.

(OR)

- b) Discuss the Laue's method in detail for the determination of crystal structure.

- 4 a) Derive the expression of de Broglie's wavelength for electron. List out the properties of matter waves.

(OR)

- b) Discuss the significance of Fermi-Dirac distribution function. Derive the function for conductors.

- 5 a) Derive the expression for the density of charge carriers in intrinsic semiconductors.

(OR)

- b) Discuss the BCS theory of superconductivity in detail.

- 6 a) Derive the expression for Lorentz field in a dielectric material.

(OR)

- b) Define Magnetic flux. Classify Dia, Para, and Ferro magnetic materials in detail.

Introduction to Programming**(Common to EEE, ECE, EBM, IT, CSE(AI), CSE(CS), CSE(IOT), CSBS, AI & AIDS)**

(For 2023, 2024 & 2025 Admitted Batches Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

1. a. Differentiate between variable and constant.
b. Define Algorithm.
c. Define break and continue with syntax.
d. Write a C program to display the numbers 1 to 10 using a do-while loop.
e. How is an array represented in memory?
f. Differentiate between the scanf () and fgets () functions.
g. Define a recursive function.
h. Define the scope of variable and its types.
i. Describe the Union with syntax.
j. List out any four file operations commands.

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Construct the flowchart to find the greatest among three numbers.
(OR)
b) Explain the top-down approach and the bottom-up approach in detail.
- 3 a) Explain if and if-else with Syntax and an Example program.
(OR)
b) Write a C program to display odd and even numbers using the for loop.
- 4 a) Write a program to do the matrix multiplication.
(OR)
b) Write a C program to reverse a given string.
- 5 a) Examine with an example program using C to display all the prime numbers between 1 to 50.
(OR)
b) Explain the difference between global variables and local variables with an example program
- 6 a) Explain nested structure and write a C Program to implement the same.
(OR)
b) Define Linked List. Write a C program to create a single-linked list and display it.

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Basic Electrical and Electronics Engineering**(Common to EEE, ECE, ECE(BM), IT, CSE(AI), CSE(CS), CSE(IOT), CSE(BS), AI & AIDS)****(For 2023, 2024 & 2025 Admitted Batches Only)**

Time: 3 Hours

Max.Marks:70

PART-A- Basic Electrical Engineering**SECTION-I****Answer all questions and all questions carry equal marks****5 x 1 = 05M**

- 1.a) Define Ohm's Law.
- 1.b) Define Power factor.
- 1.c) Write the principle of generator.
- 1.d) Mention the advantages of MI instruments.
- 1.e) Define earthing?

SECTION-II**Answer All questions and all questions carry equal marks****3 x 10 = 30M**

- 2 a) Derive an expression for Average value and RMS value in alternating quantity.
(OR)
- b) i. State and explain Kirchhoff's laws.
ii. State and explain Superposition theorem.
- 3 a) Explain the construction and working of a DC Motor with a neat sketch
(OR)
- b) Derive the expression for balance condition of Wheatstone bridge.
- 4 a) Explain the hydel power generation with neat sketch and its advantages.
(OR)
- b) i Define tariff? Explain the two-part tariff with an example.
ii. Explain MCB and its applications.

PART-B - Basic Electronics Engineering**SECTION-I****Answer all questions and all questions carry equal marks****5 x 1 = 05M**

- 5.a) What is mean by reverse bias in zener diode?
- 5.b) Draw the symbol of PNP and NPN transistor.
- 5.c) Define rectifier and mention the types of rectifier.
- 5.d) What is the use of microphone in public addressing system.
- 5.e) Draw the symbol and truth table of basic gates.

SECTION-II**Answer All questions and all questions carry equal marks****3 x 10 = 30M**

- 6 a) Describe the VI Characteristics of Zener diode with suitable examples.
(OR)
- b) Draw and explain the input and output characteristics of a BJT in Common Base Configuration with suitable examples.
- 7 a) Explain the Half wave and full wave rectifier with suitable Circuit diagram with graph. What are the applications of rectifiers.
(OR)
- b) Draw the block diagram of Regulated DC Power supply and explain each blocks.
- 8 a) Draw the circuit diagram of Full Adder and explain the operation with truth table.
(OR)
- b) What is mean by sequential circuits. Explain the operation of Basic S-R Flip flop with Diagram.

Answer all questions and all questions carry equal marks 05 x 14 = 70M

1. a) Construct a parabola whose distance of focus from directrix is 60 mm. The eccentricity is 1. Also draw a tangent and normal at 65 mm on the curve.

(OR)

- b) Draw the involute of a regular hexagon of side 30 mm. Also draw a tangent and normal to the curve at a distance 70 mm from the centre of the hexagon.

2. a) A line AB of 70mm long, has its end A at 15mm above H.P. and 10mm in front of V.P. Its front view and top view measure 50mm and 60mm respectively. Draw the projections of the line and determine its inclinations with H.P. and V.P.

(OR)

- b) A point A is 20 mm above H.P. and in the first quadrant. Its shortest distance from the reference line XY is 40mm. Draw the projections of the point and determine its distance from H.P. & V.P.

3. a) A Square pyramid, with side of base 25 mm and axis 45 mm long, is resting on a corner of its base on HP on HP. Draw the projection of the prism when its axis is making 30° with HP and parallel to VP.

(OR)

- b) A pentagonal prism of side of base 30 mm and axis 60 mm long, is resting on HP on a corner of its base. Draw the projection of the prism, when the base is inclined at 60° to HP and the axis appears to be inclined at 30° to VP.

4. a) A pentagonal pyramid, with side of base 25 mm and axis 65 mm long, is resting with its base on HP and one edge of base parallel to VP. It is cut by a sectional plane inclined to HP at 30° and passing through the axis at a point 30 mm from the apex. Draw the front view and sectional top view.

(OR)

- b) A hexagonal prism, edge of base 30 mm and axis 60 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the axis at 25 mm from its top. Develop the lateral surfaces of the truncated prism.

47
56

